MVLU COLLEGE
WSN PRACTICAL 1

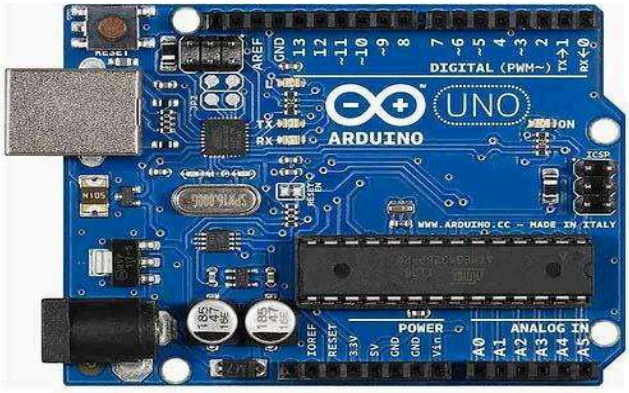
AIM:- Design a conceptual block diagram of a wireless sensor node and analyse the role of sensing unit, microcontroller, transceiver, memory, and power supply in energy-constrained environments.

1. sensing unit :- Motion Sensor (PIR Sensor)



- Detects the movement of humans or animals by sensing infrared (IR) radiation.
- Converts motion into an electrical signal for processing.
- Consumes very little power, making it suitable for wireless sensor networks.
- Activates the microcontroller only when motion is detected.
- Reduces unnecessary sensing, helping conserve battery energy.

2. Microcontroller :- Arduino Uno (ATmega328P)



A. Processes Sensor Data

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a. Receives signals from the PIR motion sensor and processes them to detect motion.

B. Controls System Operations

a. Manages the overall operation of the sensor node and controls connected devices such as LEDs, buzzers, or alarms.

C. Low Power Consumption

a. Supports sleep modes to reduce power usage, making it suitable for energy-constrained wireless sensor networks.

D. Easy Programming

a. Can be programmed using the Arduino IDE with C/C++, making it simple for students and beginners to develop applications.

E. Supports Multiple Interfaces

a. Provides digital and analog I/O pins, UART, SPI, and I²C communication for connecting sensors, wireless modules, and other peripherals.

3. Transceiver :- ESP8266 Wi-Fi Module



A. Wireless Communication

a. Sends motion detection data from the PIR sensor to a Wi-Fi network or cloud server.

B. Real-Time Alerts

a. Sends instant notifications to a mobile app or email when motion is detected.

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C. IoT Connectivity

a. Connects the PIR motion sensor to Internet of Things (IoT) platforms for remote monitoring.

D. Low Power Operation

a. Supports sleep modes to reduce power consumption in battery-powered motion sensor systems.

E. Easy Integration

a. Easily interfaces with an Arduino Uno or can be used as a standalone controller with the PIR sensor for smart home and security applications.

4. Memory :- ATmega328P



Flash Memory (32 KB)

- Stores the program code that controls the PIR motion sensor and system operation.

SRAM (2 KB)

- Temporarily stores sensor readings and variables while the program is running.

EEPROM (1 KB)

- Stores important settings such as sensitivity or configuration, even after power is turned off.

Data Buffering

- Holds motion detection data temporarily before it is transmitted through the Wi-Fi module (ESP8266).

Energy-Efficient Storage

- Efficient use of memory reduces processor activity and power consumption, increasing the battery life of the wireless motion sensor node.

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5. Power Supply (Energy-Constrained Environment) :- 18650 Li-ion battery



Provides Power

- Supplies electrical power to the PIR motion sensor, Arduino Uno, and ESP8266 Wi-Fi module.

Battery Powered

- Commonly uses a 9V battery, 18650 Li-ion battery, or 5V USB power bank for portable operation.

Supports Low Power Operation

- Works with sleep modes of the Arduino and ESP8266 to reduce energy consumption.

Ensures Long Battery Life

- Efficient power management extends the operating time of the wireless motion sensor system.

Can Use Renewable Energy

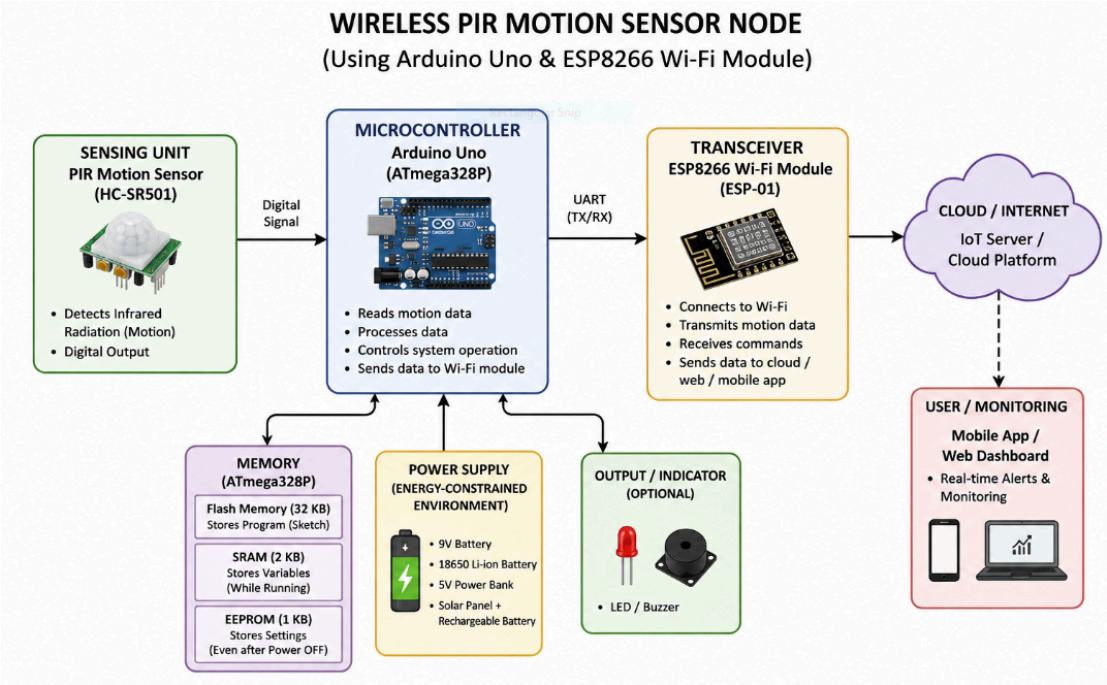
- A small solar panel with a rechargeable battery can be used for continuous operation in remote locations.

6. Block Diagram :-

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Output :-



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